

Department of CSE**ASSIGNMENT PLAN**

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Branch	Semester	Subject	Code	Session
CSE	3 rd	Computer Architecture	CCS203	Aug – Dec, 2025

ASSIGNMENT NO. 2**Course-Content Covered:**

Unit-5 Central Processing Unit and Input-Output Organization: General register organization, Stack organization, RISC and CISC architecture, I/O interface and types, Modes of transfer, DMA.

Unit-6 8085 Microprocessor Architecture and Programming: Introduction to microprocessors, 8085 microprocessor architecture Pin Diagram, Bus structure, Addressing modes, Instruction classification, Instruction formats, Data transfer operations, Arithmetic operations, Logical operations, Branch operations, Stack and subroutine operations.

Unit-7 Interfacing: 8255 Programmable Peripheral Interface, 8259 interrupt controller and 8237 DMA controller.

ASSIGNMENT QUESTIONS**Q1. Explain the following:**

- Differentiate between RISC and CISC architectures and explain how their important features impact system performance.
- Draw the block diagram of the 8085 microprocessor and analyze how its internal blocks work together with external signals shown in the 8085-pin diagram to perform a complete instruction cycle. Explain how address/data buses and control signals coordinate to fetch and execute an instruction.

Q2. Explain the following:

- Describe in detail the different types of addressing modes used in a microprocessor with suitable examples.
- Explain the step-by-step working of 8237 DMA Controller when it transfers data from an I/O device to memory without CPU involvement.